

Homework Signal 5

Week 7

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Collaborators. ChatGPT (for L^AT_EX styling and grammar checking)

1 DFT & DTFT

Problem 1. Discrete Fourier Transform (DFT):

1.1 Given the DFT spectrum $X[k]$, express the corresponding time-domain signal $x[n]$ in terms of its constituent real sinusoids.

TO SUBMIT

$$1.1.1 \quad X[0] = 3, X[1] = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}, X[2] = -2, X[3] = \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}$$

Solution. Using the inverse DFT formula:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(\frac{2\pi}{N})kn}$$

where $N = 4$.

Substituting the given values:

$$\begin{aligned} X[k] &= \left\{ 3, \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}, -2, \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}} \right\} \\ &= \left\{ 3, \cos\left(-\frac{\pi}{4}\right) + j\sin\left(-\frac{\pi}{4}\right), -2, \cos\left(\frac{\pi}{4}\right) + j\sin\left(\frac{\pi}{4}\right) \right\} \\ X[k] &= \left\{ 3, e^{-j\frac{\pi}{4}}, -2, e^{j\frac{\pi}{4}} \right\} \end{aligned}$$

Calculating $x[n]$:

$$\begin{aligned} x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j(\frac{2\pi}{N})kn} \\ &= \frac{1}{N} \left(X[0]e^{j(\frac{2\pi}{N})k(0)} + X[1]e^{j(\frac{2\pi}{N})k(1)} + X[2]e^{j(\frac{2\pi}{N})k(2)} + X[3]e^{j(\frac{2\pi}{N})k(3)} \right) \\ &= \frac{1}{4} \left(3e^{j(\frac{2\pi}{4})(0)n} + e^{-j\frac{\pi}{4}}e^{j(\frac{2\pi}{4})(1)n} - 2e^{j(\frac{2\pi}{4})(2)n} + e^{j\frac{\pi}{4}}e^{j(\frac{2\pi}{4})(3)n} \right) \\ &= \frac{1}{4} \left(3e^0 + e^{j(\frac{\pi}{4})(2n-1)} - 2e^{j\pi n} + e^{j(\frac{\pi}{4})(6n+1)-j(2\pi)} \right) \\ &= \frac{1}{4} (3 - 2(-1)^n) + \frac{1}{4} \left(e^{j(\frac{\pi}{4})(2n-1)} + e^{j(\frac{\pi}{4})(6n+1-8n)} \right) \\ &= \frac{1}{4} (3 - 2(-1)^n) + \frac{1}{4} \left(e^{j(\frac{\pi}{4})(2n-1)} + e^{-j(\frac{\pi}{4})(2n-1)} \right) \\ &= \frac{1}{4} (3 - 2(-1)^n) + \frac{1}{2} \cos\left(\frac{\pi}{4}(2n-1)\right) \\ x[n] &= \frac{1}{4} (3 - 2(-1)^n) + \frac{1}{2} \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right) \end{aligned}$$

Thus, the time-domain signal $x[n]$ is expressed as:

$$x[n] = \frac{1}{4} (3 - 2(-1)^n) + \frac{1}{2} \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right)$$

TO SUBMIT

1.1.2 $X[0] = -2, X[1] = \sqrt{3} + j, X[2] = 3, X[3] = \sqrt{3} - j$

Solution. Using the inverse DFT formula:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn}$$

where $N = 4$.

Substituting the given values:

$$\begin{aligned} X[k] &= \{-2, \sqrt{3} + j, 3, \sqrt{3} - j\} \\ &= \left\{-2, 2\left(\frac{\sqrt{3}}{2} + j\frac{1}{2}\right), 3, 2\left(\frac{\sqrt{3}}{2} - j\frac{1}{2}\right)\right\} \\ &= \left\{-2, 2\left(\cos\left(\frac{\pi}{6}\right) + j\sin\left(\frac{\pi}{6}\right)\right), 3, 2\left(\cos\left(-\frac{\pi}{6}\right) + j\sin\left(-\frac{\pi}{6}\right)\right)\right\} \\ X[k] &= \{-2, 2e^{j\frac{\pi}{6}}, 3, 2e^{-j\frac{\pi}{6}}\} \end{aligned}$$

Calculating $x[n]$:

$$\begin{aligned} x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn} \\ &= \frac{1}{N} \left(X[0] e^{j\left(\frac{2\pi}{N}\right)k(0)} + X[1] e^{j\left(\frac{2\pi}{N}\right)k(1)} + X[2] e^{j\left(\frac{2\pi}{N}\right)k(2)} + X[3] e^{j\left(\frac{2\pi}{N}\right)k(3)} \right) \\ &= \frac{1}{4} \left(-2e^{j\left(\frac{2\pi}{4}\right)(0)n} + 2e^{j\frac{\pi}{6}} e^{j\left(\frac{2\pi}{4}\right)(1)n} + 3e^{j\left(\frac{2\pi}{4}\right)(2)n} + 2e^{-j\frac{\pi}{6}} e^{j\left(\frac{2\pi}{4}\right)(3)n} \right) \\ &= \frac{1}{4} \left(-2e^0 + 2e^{j\left(\frac{\pi}{6}\right)(3n+1)} + 3e^{j\pi n} + 2e^{j\left(\frac{\pi}{6}\right)(9n-1)-j(2\pi)} \right) \\ &= \frac{1}{4} (-2 + 3(-1)^n) + \frac{1}{4} \left(2e^{j\left(\frac{\pi}{6}\right)(3n+1)} + 2e^{j\left(\frac{\pi}{6}\right)(9n-1-12n)} \right) \\ &= \frac{1}{4} (-2 + 3(-1)^n) + \frac{1}{2} \left(e^{j\left(\frac{\pi}{6}\right)(3n+1)} + e^{-j\left(\frac{\pi}{6}\right)(3n+1)} \right) \\ &= \frac{1}{4} (-2 + 3(-1)^n) + \cos\left(\frac{\pi}{6}(3n+1)\right) \\ x[n] &= \frac{1}{4} (-2 + 3(-1)^n) + \cos\left(\frac{\pi}{2}n + \frac{\pi}{6}\right) \end{aligned}$$

Thus, the time-domain signal $x[n]$ is expressed as:

$$x[n] = \frac{1}{4} (-2 + 3(-1)^n) + \cos\left(\frac{\pi}{2}n + \frac{\pi}{6}\right)$$

TO SUBMIT

1.1.3 $X[0] = 1, X[1] = 2 - j2\sqrt{3}, X[2] = -3, X[3] = 2 + j2\sqrt{3}$

Solution. Using the inverse DFT formula:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn}$$

where $N = 4$.

Substituting the given values:

$$\begin{aligned} X[k] &= \{1, 2 - j2\sqrt{3}, -3, 2 + j2\sqrt{3}\} \\ &= \left\{1, 4 \left(\frac{1}{2} - j\frac{\sqrt{3}}{2}\right), -3, 4 \left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\right\} \\ &= \left\{1, 4 \left(\cos\left(-\frac{\pi}{3}\right) + j \sin\left(-\frac{\pi}{3}\right)\right), -3, 4 \left(\cos\left(\frac{\pi}{3}\right) + j \sin\left(\frac{\pi}{3}\right)\right)\right\} \\ X[k] &= \{1, 4e^{-j\frac{\pi}{3}}, -3, 4e^{j\frac{\pi}{3}}\} \end{aligned}$$

Calculating $x[n]$:

$$\begin{aligned} x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn} \\ &= \frac{1}{N} \left(X[0] e^{j\left(\frac{2\pi}{N}\right)k(0)} + X[1] e^{j\left(\frac{2\pi}{N}\right)k(1)} + X[2] e^{j\left(\frac{2\pi}{N}\right)k(2)} + X[3] e^{j\left(\frac{2\pi}{N}\right)k(3)} \right) \\ &= \frac{1}{4} \left((1) e^{j\left(\frac{2\pi}{4}\right)(0)n} + 4e^{-j\frac{\pi}{3}} e^{j\left(\frac{2\pi}{4}\right)(1)n} + (-3) e^{j\left(\frac{2\pi}{4}\right)(2)n} + 4e^{j\frac{\pi}{3}} e^{j\left(\frac{2\pi}{4}\right)(3)n} \right) \\ &= \frac{1}{4} \left(e^0 + 4e^{j\left(\frac{\pi}{6}\right)(3n-2)} - 3e^{j\pi n} + 4e^{j\left(\frac{\pi}{6}\right)(9n+2)-j(2\pi)} \right) \\ &= \frac{1}{4} (1 - 3(-1)^n) + \frac{1}{4} \left(4e^{j\left(\frac{\pi}{6}\right)(3n-2)} + 4e^{j\left(\frac{\pi}{6}\right)(9n+2-12n)} \right) \\ &= \frac{1}{4} (1 - 3(-1)^n) + \left(e^{j\left(\frac{\pi}{6}\right)(3n-2)} + e^{-j\left(\frac{\pi}{6}\right)(3n-2)} \right) \\ &= \frac{1}{4} (1 - 3(-1)^n) + 2 \cos\left(\frac{\pi}{6}(3n-2)\right) \\ x[n] &= \frac{1}{4} (1 - 3(-1)^n) + 2 \cos\left(\frac{\pi}{2}n - \frac{\pi}{3}\right) \end{aligned}$$

Thus, the time-domain signal $x[n]$ is expressed as:

$$x[n] = \frac{1}{4} (1 - 3(-1)^n) + 2 \cos\left(\frac{\pi}{2}n - \frac{\pi}{3}\right)$$

1.2 Show that if $x[n]$ is real sequence, $X[N - k] = X^*[k]$

Solution. Using the DFT definition:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{N}\right)kn}$$

To find $X[N - k]$:

$$\begin{aligned} X[N - k] &= \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{N}\right)(N-k)n} \\ &= \sum_{n=0}^{N-1} x[n] e^{-j\left(2\pi n - \frac{2\pi}{N}kn\right)} \\ &= \sum_{n=0}^{N-1} x[n] e^{-j2\pi n} e^{j\left(\frac{2\pi}{N}\right)kn} \\ &= \sum_{n=0}^{N-1} x[n] e^{j\left(\frac{2\pi}{N}\right)kn} \quad (\text{since } e^{-j2\pi n} = 1 \text{ for integer } n) \\ &= \sum_{n=0}^{N-1} x[n] \left(e^{-j\left(\frac{2\pi}{N}\right)kn} \right)^* \\ &= \left(\sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{N}\right)kn} \right)^* \quad (\text{since } x[n] \text{ is real}) \\ X[N - k] &= X^*[k] \end{aligned}$$

Thus, we have shown that if $x[n]$ is a real sequence, then $X[N - k] = X^*[k]$.

$$\boxed{x[n] \text{ is real sequence} \implies X[N - k] = X^*[k]}$$

Problem 2. Discrete-Time Fourier Transform (DTFT):

2.1 Use the properties of the discrete-time transform to determine $X(e^{j\omega})$ for the following sequences

TO SUBMIT

2.1.1 $x[n] = \left(\frac{1}{3}\right)^{|n|}$

Solution. Using the DTFT definition:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Substituting the given sequence:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{3}\right)^{|n|} e^{-j\omega n} \\ X(e^{j\omega}) &= \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n e^{-j\omega n} + \sum_{n=-\infty}^{-1} \left(\frac{1}{3}\right)^{|n|} e^{-j\omega n} \end{aligned}$$

By substituting $m = -n$ in the second sum, we got:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{\infty} \left(\frac{1}{3}e^{-j\omega}\right)^n + \sum_{m=1}^{\infty} \left(\frac{1}{3}\right)^m e^{j\omega m} \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{3}e^{-j\omega}\right)^n + \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^m e^{j\omega m} - \left(\frac{1}{3}\right)^0 e^{j\omega(0)} \\ X(e^{j\omega}) &= \sum_{n=0}^{\infty} \left(\frac{1}{3}e^{-j\omega}\right)^n + \sum_{m=0}^{\infty} \left(\frac{1}{3}\right)^m e^{j\omega m} - 1 \end{aligned}$$

Then, using the geometric series formula $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ for $|r| < 1$:

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{1 - \frac{1}{3}e^{-j\omega}} + \frac{1}{1 - \frac{1}{3}e^{j\omega}} - 1 \\ &= \frac{1 - \frac{1}{3}e^{j\omega} + 1 - \frac{1}{3}e^{-j\omega}}{\left(1 - \frac{1}{3}e^{-j\omega}\right)\left(1 - \frac{1}{3}e^{j\omega}\right)} - 1 \\ &= \frac{\left[2 - \frac{1}{3}(e^{j\omega} + e^{-j\omega})\right] - \left[1 - \frac{1}{3}(e^{j\omega} + e^{-j\omega}) + \frac{1}{9}\right]}{1 - \frac{1}{3}(e^{j\omega} + e^{-j\omega}) + \frac{1}{9}} \\ &= \frac{2 - \frac{10}{9}}{1 - \frac{2}{3}\cos\omega + \frac{1}{9}} \\ &= \frac{\frac{8}{9}}{\frac{10}{9} - \frac{2}{3}\cos\omega} \\ X(e^{j\omega}) &= \frac{4}{5 - 3\cos\omega} \end{aligned}$$

Thus, the DTFT of the sequence $x[n] = \left(\frac{1}{3}\right)^{|n|}$ is:

$$X(e^{j\omega}) = \frac{4}{5 - 3\cos\omega}$$

TO SUBMIT

2.1.2 $x[n] = a^n \cos(\Omega_0 n) \cdot u[n]$, $|a| < 1$

Solution. Using the DTFT definition:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Substituting the given sequence:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} a^n \cos(\Omega_0 n) \cdot u[n]e^{-j\omega n} \\ X(e^{j\omega}) &= \sum_{n=0}^{\infty} a^n \cos(\Omega_0 n)e^{-j\omega n} \end{aligned}$$

Using the Euler's formula $\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$, we got:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{\infty} a^n \left(\frac{e^{j\Omega_0 n} + e^{-j\Omega_0 n}}{2} \right) e^{-j\omega n} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} a^n e^{j\Omega_0 n} e^{-j\omega n} + \frac{1}{2} \sum_{n=0}^{\infty} a^n e^{-j\Omega_0 n} e^{-j\omega n} \\ X(e^{j\omega}) &= \frac{1}{2} \sum_{n=0}^{\infty} \left(a e^{j(\Omega_0 - \omega)} \right)^n + \frac{1}{2} \sum_{n=0}^{\infty} \left(a e^{-j(\Omega_0 + \omega)} \right)^n \end{aligned}$$

Then, using the geometric series formula $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ for $|r| < 1$:

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{2} \cdot \frac{1}{1 - a e^{j(\Omega_0 - \omega)}} + \frac{1}{2} \cdot \frac{1}{1 - a e^{-j(\Omega_0 + \omega)}} \\ &= \frac{1}{2} \cdot \frac{1 - a e^{-j(\Omega_0 + \omega)} + 1 - a e^{j(\Omega_0 - \omega)}}{(1 - a e^{j(\Omega_0 - \omega)})(1 - a e^{-j(\Omega_0 + \omega)})} \\ X(e^{j\omega}) &= \frac{1}{2} \cdot \frac{2 - a(e^{j(\Omega_0 - \omega)} + e^{-j(\Omega_0 + \omega)})}{1 - a(e^{j(\Omega_0 - \omega)} + e^{-j(\Omega_0 + \omega)}) + a^2 e^{-2j\omega}} \end{aligned}$$

Consider the value of $e^{j(\Omega_0 - \omega)} + e^{-j(\Omega_0 + \omega)}$:

$$\begin{aligned} e^{j(\Omega_0 - \omega)} + e^{-j(\Omega_0 + \omega)} &= e^{j\Omega_0} e^{-j\omega} + e^{-j\Omega_0} e^{-j\omega} \\ &= e^{-j\omega} (e^{j\Omega_0} + e^{-j\Omega_0}) \\ e^{j(\Omega_0 - \omega)} + e^{-j(\Omega_0 + \omega)} &= 2e^{-j\omega} \cos \Omega_0 \end{aligned}$$

Substituting back:

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{2} \cdot \frac{2 - 2a e^{-j\omega} \cos \Omega_0}{1 - 2a e^{-j\omega} \cos \Omega_0 + a^2 e^{-2j\omega}} \\ X(e^{j\omega}) &= \frac{1 - a e^{-j\omega} \cos \Omega_0}{1 - 2a e^{-j\omega} \cos \Omega_0 + a^2 e^{-2j\omega}} \end{aligned}$$

Thus, the DTFT of the sequence $x[n] = a^n \cos(\Omega_0 n) \cdot u[n]$ is:

$$X(e^{j\omega}) = \frac{1}{2} \left(\frac{1}{1 - a e^{j(\Omega_0 - \omega)}} + \frac{1}{1 - a e^{-j(\Omega_0 + \omega)}} \right) = \frac{1 - a e^{-j\omega} \cos \Omega_0}{1 - 2a e^{-j\omega} \cos \Omega_0 + a^2 e^{-2j\omega}}$$

TO SUBMIT

2.1.3 $x[n] = (n+1)a^n \cdot u[n]$, $|a| < 1$

Solution. Using the Linearity property of DTFT:

$$X(e^{j\omega}) = \text{DTFT} \{(n+1)a^n \cdot u[n]\} = \text{DTFT} \{na^n \cdot u[n]\} + \text{DTFT} \{a^n \cdot u[n]\}$$

Define $y[n] = a^n \cdot u[n]$, and $z[n] = ny[n]$, we get:

$$X(e^{j\omega}) = \text{DTFT} \{z[n]\} + \text{DTFT} \{y[n]\}$$

From the property of DTFT for $y[n] = a^n \cdot u[n]$ $|a| < 1$:

$$Y(e^{j\omega}) = \text{DTFT} \{y[n]\} = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \frac{1}{1 - ae^{-j\omega}}$$

Using the Differentiation property of DTFT:

$$\text{DTFT} \{nx[n]\} = j \frac{dX(e^{j\omega})}{d\omega}$$

where $X(e^{j\omega}) = \text{DTFT} \{x[n]\}$.

Thus, we have:

$$\begin{aligned} Z(e^{j\omega}) &= j \frac{dY(e^{j\omega})}{d\omega} \\ &= j \frac{d}{d\omega} \left(\frac{1}{1 - ae^{-j\omega}} \right) \\ &= j \cdot \left(\frac{0 \cdot (1 - ae^{-j\omega}) - 1 \cdot (-a(-j)e^{-j\omega})}{(1 - ae^{-j\omega})^2} \right) \\ &= j \cdot \left(\frac{-jae^{-j\omega}}{(1 - ae^{-j\omega})^2} \right) \\ Z(e^{j\omega}) &= \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2} \end{aligned}$$

Combining both results:

$$\begin{aligned} X(e^{j\omega}) &= Z(e^{j\omega}) + Y(e^{j\omega}) \\ &= \frac{ae^{-j\omega}}{(1 - ae^{-j\omega})^2} + \frac{1}{1 - ae^{-j\omega}} \\ &= \frac{ae^{-j\omega} + (1 - ae^{-j\omega})}{(1 - ae^{-j\omega})^2} \\ X(e^{j\omega}) &= \frac{1}{(1 - ae^{-j\omega})^2} \end{aligned}$$

Thus, the DTFT of the sequence $x[n] = (n+1)a^n \cdot u[n]$ is:

$$X(e^{j\omega}) = \frac{1}{(1 - ae^{-j\omega})^2}$$